

SIDDH MANDIRWALA

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EDUCATION

- New York University**, M.S. Computer Science Sep 2024 – May 2026
- GPA 3.815 / 4.0; coursework in Algorithms, AI, Machine Learning, Big Data, Software Engineering, Information Security & Privacy. Student Engagement Ambassador (raised \$35,000 for scholarships).
- Gujarat Technological University**, B.E. Computer Engineering Sep 2020 – Jun 2024
- Grade 9.20 / 10.0; coursework in Data Structures & Algorithms, OOP, Computer Networks, and Cloud Computing.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, Java, C/C++, SQL

Frontend: Next.js, React.js, Vue.js, Tailwind CSS, HTML5, CSS3, Framer Motion, D3.js

Backend & APIs: Node.js, Express.js, Flask, Strapi, GraphQL (Apollo), REST APIs, WebSockets, NextAuth, JWT, Stripe

Databases: PostgreSQL, MySQL, MongoDB, Redis, Qdrant

DevOps & Cloud: Git/GitHub, Docker, Docker Compose, Kubernetes (K8s), Vercel, Railway, AWS, CI/CD

Concepts & Practices: Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design, Microservices, RESTful APIs, Agile/Scrum, Unit Testing, Version Control

EXPERIENCE

- AI4Purpose**, Software Engineer Intern | New York, NY Feb 2026 – May 2026
- Shipped production features for a **multi-tenant** retail platform that runs a direct-to-consumer storefront and a B2B marketplace out of one **Next.js 16** / **React 19** / **TypeScript** codebase, backed by **Strapi v5** and **PostgreSQL**.
 - Built storefront pages on the **Next.js App Router** with **React Server Components** and streaming **Suspense** over an **Apollo GraphQL** layer, running data queries in parallel and streaming each route in behind a loading skeleton.
 - Created the marketplace's price-comparison view, which pivots **754 real distributor-priced products** across a four-month window, and scoped every query to the signed-in vendor on the server so no tenant ever sees another's pricing.
 - Integrated **Stripe** subscriptions and **Stripe Connect** payouts, pricing carts on the server rather than trusting the client and decrementing stock in raw **SQL transactions** so two buyers can never claim the same unit.
 - Secured sign-in for four role-based portals with **NextAuth/Auth.js**, giving each portal its own **JWT** secret, keeping tokens off the client, and writing privileged actions to an append-only audit log hashed with **SHA-256**.
- KeyToZ**, Software Development Engineer Intern | Surat, India Jul 2023 – Jul 2024
- Worked across the stack of a healthcare web platform, building interfaces in **Vue.js** and **React.js** and services in **Node.js** and **Strapi** over **PostgreSQL**, exposed through **GraphQL** and **RESTful APIs**.
 - Built PostMinder, a post-scheduling tool with a **Vue.js + Vite** frontend, an **Express.js** REST API over PostgreSQL, and a **BullMQ + Redis** job queue so every post publishes at its exact scheduled time, all run through **Docker Compose**.
 - Traced production bugs from the browser down to the database, tuned slow queries and **REST** endpoints to bring response times down, and repaired mobile layouts, working in **Agile** sprints alongside designers and backend engineers.

PROJECTS

Parachute [GitHub](#)

- Tech:** TypeScript, Next.js 16, React 19, Node.js, WebSockets, Yjs CRDTs, Monaco Editor, xterm.js, node-pty.
- Built a browser-based **collaborative IDE** (team of three, YHack 2026) pairing a code editor with a live shared terminal so developers and their **AI agents** build together with live cursors; deployed on **Fly.io**.
- Kept simultaneous edits conflict-free with **CRDTs** and wrote a two-way sync between the shared document and the filesystem, using **chokidar**, a **300ms write debounce**, and **MD5** hashing to stop echo loops.
- Added an orchestrator that hands each agent a **file-level lock** before it writes, so two agents never edit the same file into a merge conflict.

RAGStack [GitHub](#)

- Tech:** Python, Flask, FastAPI, MongoDB, Qdrant, Docker, GPT-2, LoRA (PEFT), Hugging Face, ClearML, Gradio.
- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system: custom scrapers pull GitHub, Medium, and LinkedIn into a document store, with **top-5 cosine retrieval** served over **384-dimensional MiniLM** embeddings.
- Fine-tuned the language model with low-rank adapters on an **8-bit-quantized** base over **50 epochs** using **3,200+** curated Q&A pairs, published the weights publicly, and served everything as five services with every inference logged.

Vizpromax [GitHub](#)

- Tech:** D3.js, Next.js, TypeScript, React, Vercel.
- Turned **7+ million** NYPD complaint records (2015–2024) into a guided visual story with **8 interactive visualizations**, including a hand-built **Sankey** and a precinct-level **choropleth**, querying NYC Open Data live via server-side **SoQL**.